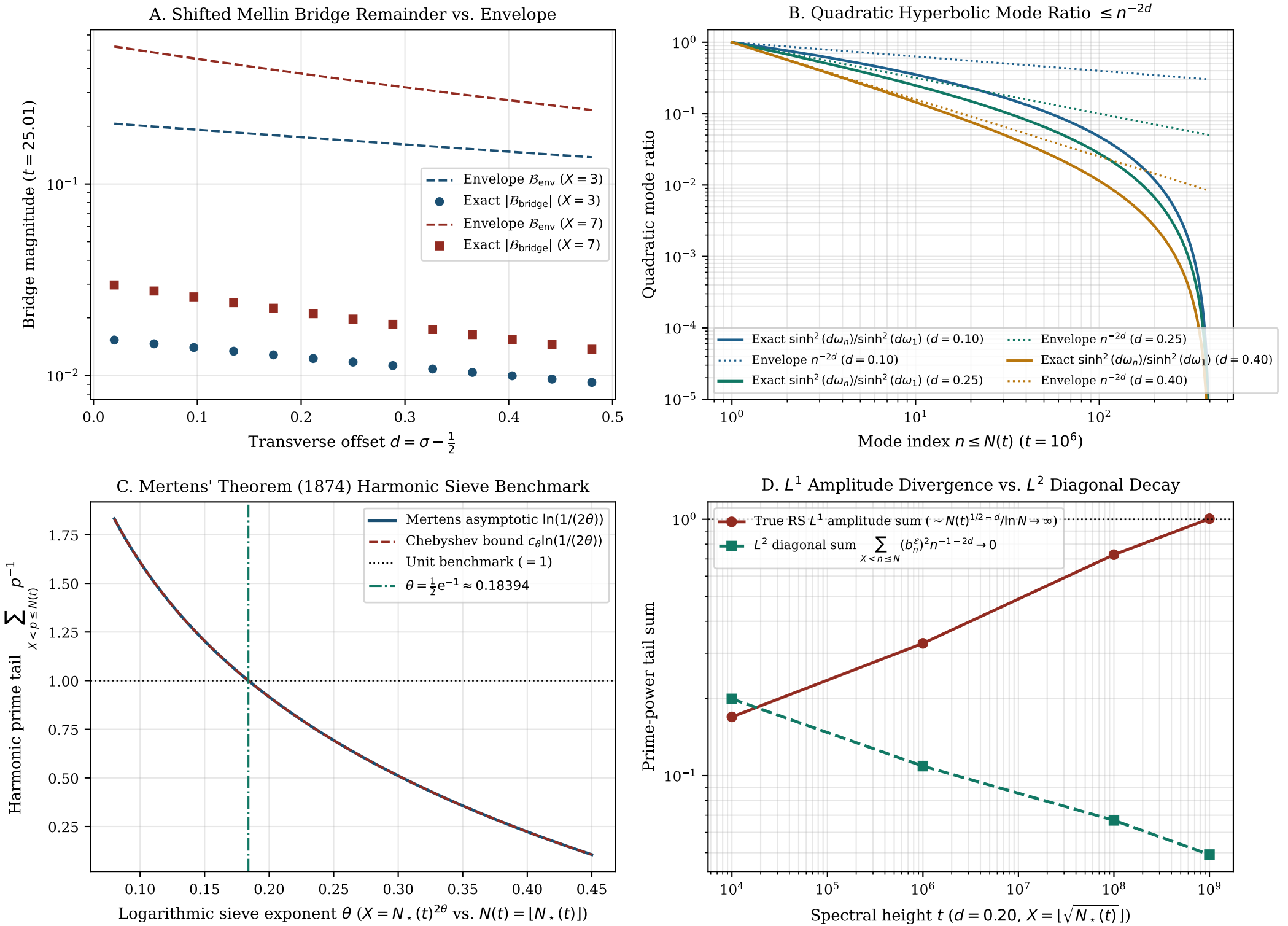

A. Shifted Mellin Bridge Remainder vs. Envelope
Bridge magnitude ($t = 25.01$)
Transverse offset $d = \sigma - \frac{1}{2}$
Envelope $\mathcal{B}_{\text{env}}$ ($X = 3$)
Exact $|\mathcal{B}_{\text{bridge}}|$ ($X = 3$)
Envelope $\mathcal{B}_{\text{env}}$ ($X = 7$)
Exact $|\mathcal{B}_{\text{bridge}}|$ ($X = 7$)
B. Quadratic Hyperbolic Mode Ratio $\leq n^{-2d}$
Quadratic mode ratio
Mode index $n \leq N(t)$ ($t = 10^6$)
Exact $\sinh^2(d\omega_n)/\sinh^2(d\omega_1)$ ($d = 0.10$)
Envelope n^{-2d} ($d = 0.10$)
Exact $\sinh^2(d\omega_n)/\sinh^2(d\omega_1)$ ($d = 0.25$)
Envelope n^{-2d} ($d = 0.25$)
Exact $\sinh^2(d\omega_n)/\sinh^2(d\omega_1)$ ($d = 0.40$)
Envelope n^{-2d} ($d = 0.40$)
C. Mertens' Theorem (1874) Harmonic Sieve Benchmark
Harmonic prime tail $\sum_{X < p \leq N(t)} p^{-1}$
Logarithmic sieve exponent θ ($X = N_\star(t)^{2\theta}$ vs. $N(t) = \lfloor N_\star(t) \rfloor$)
Mertens asymptotic $\ln(1/(2\theta))$
Chebyshev bound $c_\vartheta \ln(1/(2\theta))$
Unit benchmark ($= 1$)
$\theta = \frac{1}{2}e^{-1} \approx 0.18394$
D. L^1 Amplitude Divergence vs. L^2 Diagonal Decay
Prime-power tail sum
Spectral height t ($d = 0.20$, $X = \lfloor \sqrt{N_\star(t)} \rfloor$)
True RS L^1 amplitude sum ($\sim N(t)^{1/2-d}/\ln N \to \infty$)
L^2 diagonal sum $\sum_{X < n \leq N} (b_n^{\mathcal{E}})^2 n^{-1-2d} \to 0$